

SC1007 Data Structures & Algorithms
NTU BSc Computer Science — Comprehensive Textbook (0–100)

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Chapter 1

Arrays, Linked Lists & Foundations

“The machine can only store bits contiguously or linked by pointers.” — Every data structure trades off these two extremes.

Prerequisites: SC1003 (C/Python basics). This chapter is the 0→1: how data lives in memory and how STL wraps it.

Learning Outcomes

After this chapter you will be able to:

- (L1) compare array vs. linked-list layouts and their $O(\cdot)$ costs;
- (L2) implement singly/doubly linked lists with invariants;
- (L3) analyse dynamic arrays (amortised doubling) and use STL `vector`;
- (L4) choose the right sequential container and justify it.

1.1 Contiguous Storage: Arrays

A *pointer* is a variable whose value is a memory address; dereferencing follows the address to the stored object. Asymptotic costs: $f(n) = \Theta(g(n))$ means $c_1g(n) \leq f(n) \leq c_2g(n)$ for large n (tight bound), $O(g)$ is an upper bound, $\Omega(g)$ a lower bound. $O(1)$ = constant time independent of n ; $O(n)$ = linear in n .

An *array* stores n elements consecutively. If base address is B and each element occupies w bytes,

$$\text{addr}(A[i]) = B + i \cdot w, \tag{1.1}$$

so random access is $\Theta(1)$. Insertion/deletion at position k shifts $n - k$ elements: $\Theta(n)$.

Static vs. dynamic. Static arrays have fixed capacity. A *dynamic array* (C++ `vector`, Java `ArrayList`, Python `list`) doubles capacity when full. Appending n items costs $O(n)$ total — amortised $O(1)$ per append.

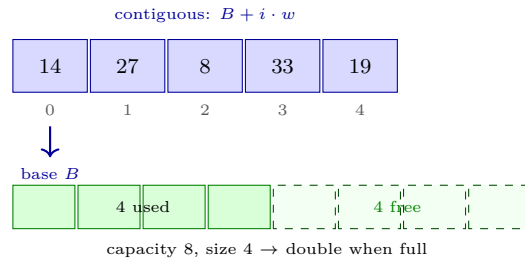


Figure 1.1: Array in memory (top): index $\rightarrow$ address in $O(1)$. Dynamic array (bottom): doubling gives amortised $O(1)$ append.

```

1 // C++ STL vector -- dynamic array
2 #include <vector>
3 std::vector<int> v = {14, 27, 8};
4 v.push_back(33);           // amortised O(1)
5 int x = v[1];              // O(1) random access
6 v.insert(v.begin()+1, 99); // O(n) -- shifts right

```

1.2 Linked Storage: Linked Lists

A *linked list* scatters nodes; each node holds data plus pointer(s) to neighbours. No shifting is needed for mid-list insertion.

Definition 1.1 (Singly linked list). A chain $p_0 \rightarrow p_1 \rightarrow \dots \rightarrow p_{n-1} \rightarrow \text{null}$ where node p_i stores (x_i, next) . Doubly linked adds prev.

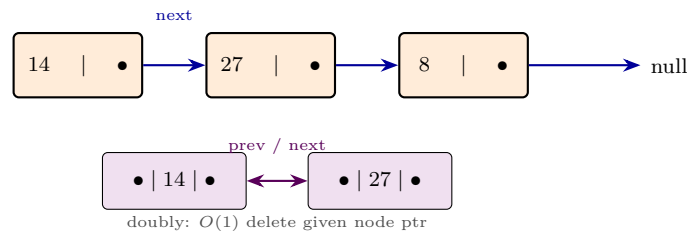


Figure 1.2: Singly linked list (top) vs. doubly linked fragment (bottom). Pointer chasing replaces index arithmetic.

Operations: access by index $\Theta(n)$ (must walk); insert/delete at head $\Theta(1)$; after locating position k , pointer rewiring is $\Theta(1)$. Doubly linked allows $O(1)$ deletion given a node pointer.

```

1 # Linked list node -- Python
2 class Node:
3     def __init__(self, val, nxt=None):
4         self.val = val
5         self.next = nxt
6
7     def insert_after(self, node, val):
8         node.next = Node(val, node.next) # O(1)
9
10    def delete_after(self, node):

```

```

11     if node.next:
12         node.next = node.next.next    # O(1)

```

1.3 Complexity Comparison & STL Choice

Operation	Array / vector	Linked list
Access $A[k]$	$\Theta(1)$	$\Theta(n)$
Append at end	amort. $\Theta(1)$	$\Theta(1)^*$
Insert at k	$\Theta(n-k)$ shift	$\Theta(k)$ seek + $\Theta(1)$ link
Delete at k	$\Theta(n-k)$ shift	$\Theta(k)$ seek + $\Theta(1)$ link
Memory overhead	low (contiguous)	+1-2 ptrs / elem

Table 1.1: Sequential containers compared. *With tail pointer; else $\Theta(n)$.

Rule of thumb: prefer **vector** unless you need frequent mid-list splicing with iterators or non-contiguous growth without reallocation.

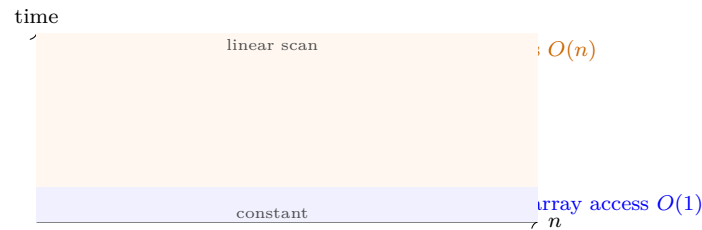


Figure 1.3: Random access: array stays flat; linked list grows linearly with n .

1.4 Iterators & Amortised Doubling

STL decouples containers from algorithms via *iterators*. A vector iterator is a random-access pointer; a list iterator is bidirectional. Algorithms like `std::sort` require random access, so they work on vectors but not lists.

Amortised doubling proof. Start with capacity 1. Insert n items, doubling when full. Total copies:

$$1 + 2 + 4 + \dots + 2^{\lceil \log_2 n \rceil - 1} < 2n. \quad (1.2)$$

So n appends cost $O(n)$ and average $O(1)$. Halving when size $<$ capacity/4 keeps both push and pop amortised $O(1)$ without thrashing.

```

1  // Iterator example -- same code works for vector and deque
2  #include <vector>
3  #include <algorithm>
4  std::vector<int> v = {5,2,9,1,7};
5  std::sort(v.begin(), v.end()); // requires random-access iterators
6  for (auto it = v.begin(); it != v.end(); ++it) *it += 1;

```

Remark 1.1. Python list over-allocates similarly; CPython grows by $\approx 1.125\times$ for large lists. Java `ArrayList` grows by $1.5\times$. Constants affect memory but not the $O(1)$ amortised bound.

1.5 Chapter Notes

CLRS Ch. 10, Weiss Ch. 3. STL: `vector` is dynamic array; `list` is doubly linked; `deque` is in Ch. 2. Python `list` is a dynamic array, not a linked list.

1.6 Exercises

- E1.1 **Address arithmetic.** Array `int A[100]` at base $B = 0x1000$, $w = 4$ bytes. (a) Address of `A[37]`? (b) Why does `A[i][j]` in $m \times n$ row-major cost $B + (i \cdot n + j)w$?
- E1.2 **Doubling analysis.** Starting from capacity 1, insert $n = 2^k$ items with doubling. Count total element copies. Show it is $< 2n$ and deduce amortised $O(1)$.
- E1.3 **Linked list reversal.** Write iterative $O(n)$ -time $O(1)$ -space reversal for singly linked list. Prove your loop invariant: after t iterations the prefix of length t is reversed.
- E1.4 **Container choice.** For each scenario choose `vector` vs. `list` and justify: (a) 1M insertions at random positions with iterator kept; (b) binary search over sorted data; (c) queue with push/pop at ends only.
- E1.5 **Splice.** Given doubly linked list with head/tail, splice node x (in L_1) into L_2 after node y in $O(1)$. Give pointer updates and handle head/tail edge cases.

Chapter 2

Stacks, Queues & Deques

“*Push, pop, enqueue, dequeue.*” — Four operations, yet they power recursion, BFS, undo, and scheduling.

Prerequisites: Ch. 1 (arrays/lists, $O(\cdot)$). This chapter adds restricted-access structures and classic applications.

Learning Outcomes

After this chapter you will be able to:

- (L1) implement stacks/queues/deques via arrays and linked lists in $O(1)$;
- (L2) use stacks for matching, postfix evaluation, and recursion simulation;
- (L3) use queues for BFS preview and scheduling; explain circular buffers;
- (L4) choose among stack / queue / deque and analyse amortised costs.

2.1 Stacks: LIFO

A *stack* is LIFO: **push** adds at the top, **pop** removes from the top, **peek** inspects. All $\Theta(1)$.

```
1 # Stack via Python list -- amortised  $O(1)$  push/pop
2 stack = []
3 stack.append(33)    # push
4 stack.append(27)
5 x = stack.pop()     # pop -> 27
6 top = stack[-1]     # peek -> 33
```

Applications: balanced brackets (push open, pop on close), postfix evaluation ($3\ 4\ + \rightarrow$ push 3, push 4, pop twice, push 7), call stack (each call pushes a frame), DFS via explicit stack.

```
1 /* Balanced brackets -- C */
```

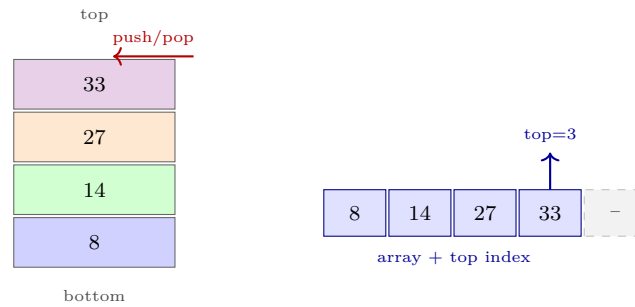


Figure 2.1: Stack plates (LIFO) and array implementation with **top** pointer. Linked-list version uses **head** as **top**.

```

2  int balanced(const char *s) {
3      char st[1000]; int top = -1;
4      for (int i = 0; s[i]; i++) {
5          if (s[i]=='(' || s[i]=='[') st[++top] = s[i];
6          else if (s[i]==')') { if (top<0 || st[top--]!='(') return 0; }
7          else if (s[i]==']') { if (top<0 || st[top--]!='[') return 0; }
8      }
9      return top == -1;
10 }

```

2.2 Queues: FIFO

A *queue* is FIFO: **enqueue** at rear, **dequeue** from front, both $\Theta(1)$ with a linked list (head/tail) or a *circular buffer* over an array.

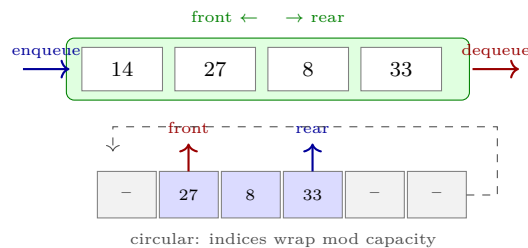


Figure 2.2: Queue (FIFO) and circular-buffer array implementation with **front**/**rear** wrapping.

Circular buffer: store **front**, **size**, **cap**; **rear** = (**front** + **size**) mod **cap**. Enqueue/dequeue adjust pointers modulo **cap**; no shifting.

2.3 Deques

A *deque* (double-ended queue) allows push/pop at *both* ends in $\Theta(1)$. Implemented by doubly linked list or a circular buffer with two pointers. C++ **deque** gives $O(1)$ random access plus $O(1)$ ends.

```

1  from collections import deque
2  dq = deque([27, 8])
3  dq.appendleft(14)    # O(1) left push

```

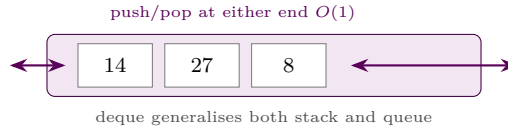


Figure 2.3: Deque supports operations at both ends, subsuming stack and queue.

```

4 dq.append(33)           # O(1) right push
5 dq.popleft()           # O(1) -> 14
6 dq.pop()               # O(1) -> 33

```

Use cases: sliding-window maximum (monotone deque, Ch. 8), work stealing, palindrome checking, undo/redo with two stacks.

2.4 Implementations Compared

Structure	Array	Linked list	Circular buf.
Stack push/pop	$O(1)$	$O(1)$	—
Queue enq/deq	$O(n)$ shift	$O(1)$	$O(1)$
Deque both ends	$O(n)$	$O(1)$	$O(1)$
Cache locality	excellent	poor	excellent

Table 2.1: Implementation trade-offs. Circular buffer wins for queue/deque on arrays.

2.5 Applications: Matching, Evaluation, BFS Preview

Postfix evaluation. Scan tokens left to right: push operands; on operator pop two, apply, push result. Example $3\ 4\ +\ 2\ \times\ =\ 14$. Infix $\rightarrow$ postfix via shunting-yard (operator stack) handles precedence.

Function calls. The call stack stores frames (locals, return address). Recursion depth = stack height; tail recursion can be optimised to iteration.

BFS preview. A queue gives level-order traversal (Ch. 3) and shortest paths in unweighted graphs (Ch. 7). Replacing the queue with a stack yields DFS — same code, different data structure, radically different order.

```

1 # Postfix evaluator -- O(n)
2 def eval_postfix(tokens):
3     st = []
4     for tok in tokens:
5         if tok in "+-*/":
6             b, a = st.pop(), st.pop()
7             st.append(str(eval(f"{a}{tok}{b}")))
8         else:
9             st.append(tok)
10    return int(st[0])
11 # eval_postfix(["3","4","+","2","*"]) == 14

```

Remark 2.1. Monotone deque for sliding-window maximum: maintain decreasing deque of candidates; each element enters and leaves once — $O(n)$ for window size k over n elements, previewing Ch. 8.

2.6 Chapter Notes

CLRS Ch. 10.1, Weiss Ch. 3.3–3.6. STL: `stack` and `queue` are adapters over `deque` by default; `priority_queue` is in Ch. 5.

2.7 Exercises

- E2.1 **Circular queue.** Implement queue with array $A[0..cap-1]$, integers `front`, `sz`. Give `enqueue`, `dequeue`, `isEmpty`, `isFull` in $O(1)$. How do you distinguish full vs. empty?
- E2.2 **Two stacks, one array.** Use a single array of size n for two stacks growing from opposite ends. Give `push/pop` for each and prove overflow iff total items $> n$.
- E2.3 **Postfix evaluation.** Evaluate $5\ 1\ 2\ +\ 4\ *\ +\ 3\ -$ with a stack, showing contents after each token. Write $O(n)$ algorithm for general postfix over $+, -, \times, /$.
- E2.4 **Queue via stacks.** Implement queue using two stacks with amortised $O(1)$ `enqueue/dequeue`. Prove amortised bound via potential $\Phi = \text{size of input stack}$.
- E2.5 **Deque min.** Design deque supporting `pushFront/Back`, `popFront/Back`, and `getMin` in $O(1)$ amortised (hint: auxiliary monotone deque).

Chapter 3

Trees & Traversals

“A tree is a hierarchy you can walk without getting lost.” — Traversals turn branching structure into linear order.

Prerequisites: Ch. 1–2 (pointers, recursion, stack/queue). This chapter is the foundation for all search trees (Ch. 4–5) and graph traversals (Ch. 7).

Learning Outcomes

After this chapter you will be able to:

- (L1) define trees, binary trees, height/depth and prove n nodes $\rightarrow n - 1$ edges;
- (L2) implement the three DFS traversals (pre/in/postorder) recursively and iteratively;
- (L3) perform level-order (BFS) traversal and relate it to queues;
- (L4) reconstruct a binary tree from traversal pairs and analyse $O(n)$ costs.

3.1 Tree Terminology

A *tree* is a connected acyclic graph. A *rooted tree* designates one node as root; every other node has a unique parent. A *binary tree* gives each node at most two children: **left** and **right** (order matters).

For node v : *depth* = edges from root to v ; *height* = longest downward path to a leaf. Height of tree = height of root. Height of single-node tree = 0; empty tree = -1 by convention.

Theorem 3.1. A tree with n nodes has exactly $n - 1$ edges. A binary tree of height h has at most $2^{h+1} - 1$ nodes.

Storage: each node holds **val**, **left**, **right** pointers. Array embedding (heap-style, Ch. 5) stores node i 's children at $2i, 2i + 1$.

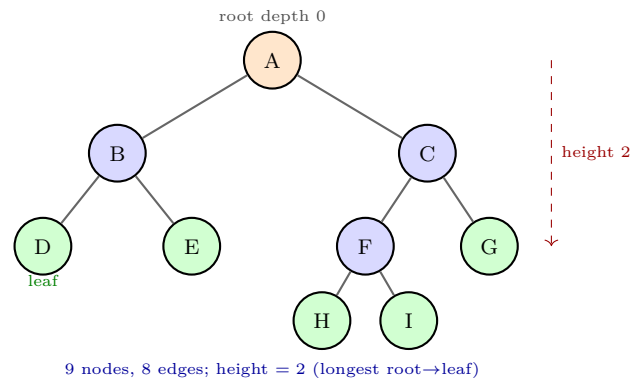


Figure 3.1: A rooted binary tree. Orange = root, blue = internal, green = leaves. Depth and height illustrated.

3.2 Depth-First Traversals

DFS visits recursively; three orders differ only in when the root is processed:

Preorder: root → left → right

Inorder: left → root → right

Postorder: left → right → root

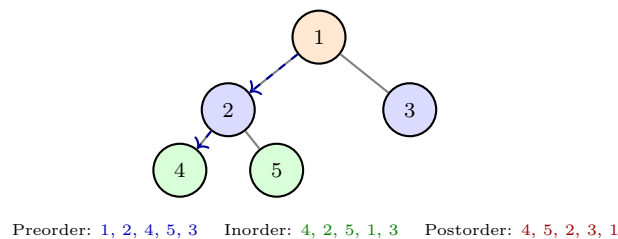


Figure 3.2: Five-node example with all three DFS orders. Preorder is used to copy; postorder to delete; inorder gives sorted order for BSTs (Ch. 4).

```

1  # Recursive traversals -- O(n) time, O(h) stack space
2  def preorder(root):
3      if not root: return
4      visit(root)           # root first
5      preorder(root.left)
6      preorder(root.right)
7
8  def inorder(root):
9      if not root: return
10     inorder(root.left)
11     visit(root)           # root in middle
12     inorder(root.right)
13
14  def postorder(root):
15      if not root: return
16      postorder(root.left)
17      postorder(root.right)
18      visit(root)         # root last

```

```

19
20 # Iterative preorder with explicit stack
21 def preorder_iter(root):
22     stack = [root] if root else []
23     while stack:
24         v = stack.pop()
25         visit(v)
26         if v.right: stack.append(v.right)
27         if v.left:  stack.append(v.left)

```

Each node is visited once, each edge traversed twice (down and up): $\Theta(n)$ time, $\Theta(h)$ stack space ($\Theta(n)$ worst case for a chain, $\Theta(\log n)$ for balanced).

3.3 Level Order (BFS)

Level order visits nodes breadth-first using a queue:

```

1 from collections import deque
2 def level_order(root):
3     if not root: return
4     q = deque([root])
5     while q:
6         v = q.popleft()
7         visit(v)
8         if v.left: q.append(v.left)
9         if v.right: q.append(v.right)

```

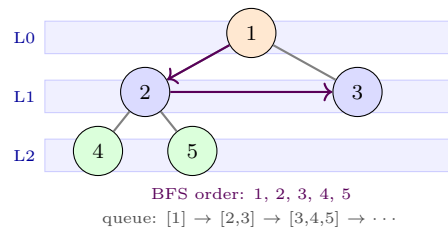


Figure 3.3: Level-order traversal processes level d before $d + 1$. Queue holds the frontier.

3.4 Reconstruction

Given inorder + preorder (or inorder + postorder) the tree is uniquely determined. Preorder gives root first; inorder splits left/right subtrees. $O(n)$ with a hash map from value to inorder index. Preorder + postorder alone is ambiguous for general binary trees.

3.5 Tree Properties & Variations

Full / complete / perfect. Full: every node has 0 or 2 children. Complete: all levels full except last, filled left-to-right (heap shape, Ch. 5). Perfect: all leaves at same depth, $n = 2^{h+1} - 1$.

Threaded trees. Replace null child pointers with threads to inorder successor/predecessor, enabling $O(1)$ -space inorder without stack.

Expression trees. Leaves are operands, internal nodes operators. Inorder with parentheses recovers infix; preorder gives prefix (Polish); postorder gives postfix — directly linking traversals to Ch. 2 evaluation.

```

1  # Iterative inorder -- O(n) time, O(h) space, no recursion
2  def inorder_iter(root):
3      stack, cur, out = [], root, []
4      while stack or cur:
5          while cur:
6              stack.append(cur); cur = cur.left
7          cur = stack.pop()
8          out.append(cur.val)
9          cur = cur.right
10     return out

```

3.6 Chapter Notes

CLRS Ch. 10.4, 12. Weiss Ch. 4. DFS $\leftrightarrow$ stack, BFS $\leftrightarrow$ queue duality previews Ch. 7 graphs.

3.7 Exercises

E3.1 **Orders.** For the tree in Fig. 3.1, list preorder, inorder, postorder, and level order.

E3.2 **Iterative inorder.** Give an $O(n)$ iterative inorder using one stack (no recursion). Prove it visits nodes in inorder and uses $O(h)$ space.

E3.3 **Reconstruction.** Preorder = [3, 9, 20, 15, 7], inorder = [9, 3, 15, 20, 7]. Reconstruct the tree and give its postorder. Explain the $O(n)$ hash-map algorithm.

E3.4 **Height vs. nodes.** Prove: a binary tree with n nodes has height $h \geq \lceil \log_2(n+1) \rceil - 1$, with equality for perfect trees. Give an example where $h = n - 1$.

E3.5 **Expression tree.** Build the tree for $(3 + 4) \times (5 - 2)$ and show that preorder/postorder/inorder give prefix/postfix/infix notation. Which needs parentheses?

Chapter 4

Binary Search Trees & Balanced Trees

“A sorted tree is a sorted array you can grow.” — But only if it stays balanced.

Prerequisites: Ch. 3 (trees, traversals). BST is the bridge from trees to ordered dictionaries; AVL/red-black keep it $O(\log n)$.

Learning Outcomes

After this chapter you will be able to:

- (L1) maintain the BST invariant and implement search/insert/delete in $O(h)$;
- (L2) explain why worst-case $h = \Theta(n)$ and how balancing fixes it;
- (L3) perform AVL rotations and verify the AVL balance invariant;
- (L4) sketch red-black invariants and argue $O(\log n)$ height.

4.1 Binary Search Trees

Definition 4.1 (BST invariant). For every node v , all keys in $v.\text{left}$ are $< v.\text{key}$ and all keys in $v.\text{right}$ are $> v.\text{key}$ (assume distinct keys).

Consequence: inorder traversal yields keys in sorted order (Ch. 3).

Operations (all $O(h)$ where h is height):

```
1 def search(root, k):
2     while root and root.key != k:
3         root = root.left if k < root.key else root.right
4     return root # None if absent
5
6 def insert(root, k):
7     if not root: return Node(k)
```

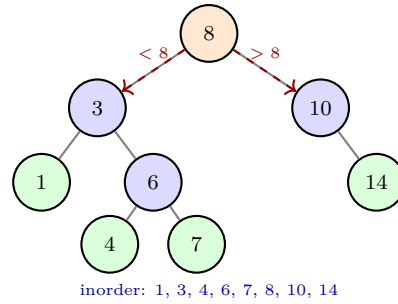


Figure 4.1: A BST. Every left subtree holds smaller keys, every right subtree larger. Inorder is sorted.

```

8   if k < root.key: root.left = insert(root.left, k)
9   elif k > root.key: root.right = insert(root.right, k)
10  return root
11
12 def delete(root, k): # three cases: leaf / one child / two children
13     if not root: return None
14     if k < root.key: root.left = delete(root.left, k)
15     elif k > root.key: root.right = delete(root.right, k)
16     else: # found
17         if not root.left: return root.right
18         if not root.right: return root.left
19         # two children: replace with successor
20         succ = min_node(root.right)
21         root.key = succ.key
22         root.right = delete(root.right, succ.key)
23     return root

```

Successor = minimum in right subtree (leftmost descendant). All ops follow one root-to-leaf path: $O(h)$ time, $O(h)$ stack space.

Degeneracy. Inserting sorted keys $1, 2, 3, \dots, n$ yields a chain with $h = n - 1$: every operation degrades to $\Theta(n)$. Balanced trees prevent this.

4.2 AVL Trees

AVL (Adelson-Velsky–Landis, 1962) maintains the *AVL invariant*: for every node,

$$|\text{height}(\text{left}) - \text{height}(\text{right})| \leq 1. \quad (4.1)$$

This forces $h = \Theta(\log n)$ (Fibonacci bound: $n \geq F_{h+2} - 1$, so $h \leq 1.44 \log_2 n$).

Rebalancing via rotations. After insertion/deletion, walk up updating heights; at the first unbalanced node, apply one of four rotations:

Insert/delete may need $O(\log n)$ rotations walking to root, but each rotation is $O(1)$.

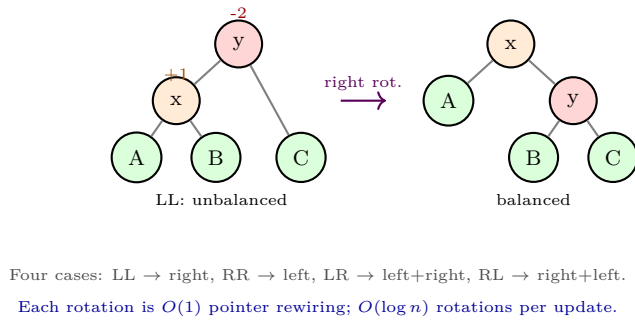


Figure 4.2: AVL right rotation for the LL case (left child too tall). The other three cases are symmetric. After rotation, heights of x, y are recomputed and the invariant is restored.



Figure 4.3: Height comparison: unbalanced BST can be linear; AVL guarantees logarithmic height.

4.3 Red-Black Trees (Sketch)

Red-black trees relax the AVL condition for fewer rotations. Each node is coloured red or black with invariants: (i) root black, (ii) no two reds adjacent, (iii) every root-to-leaf path has same number of black nodes (black-height). These imply $h \leq 2 \log_2(n + 1)$.

Insertion colours the new node red, then fixes violations by recolouring and at most two rotations along the path. STL `map/set` (C++) and Java `TreeMap` are typically red-black trees.

AVL is more strictly balanced (faster lookups); red-black has fewer rotations on updates (faster insertions). Both are $O(\log n)$ worst-case.

4.4 Analysis & Rotation Details

Search cost = $h + 1$ comparisons. In a random BST built from random insertions, expected $h = \Theta(\log n)$; worst-case $h = n - 1$ (sorted input). This motivates self-balancing.

Rotations preserve inorder. A right rotation at y with left child x :

$$\text{inorder}(y \text{ subtree before}) = A, x, B, y, C = \text{inorder}(x \text{ subtree after}), \quad (4.2)$$

where A, B, C are subtrees. Only $O(1)$ pointers change; heights of x, y are recomputed. Left rotation is symmetric; LR = left on x then right on y .

1 `# Right rotation at y (y.left == x) -- O(1)`

```

2 | def rotate_right(y):
3 |     x = y.left
4 |     y.left = x.right
5 |     x.right = y
6 |     update_height(y); update_height(x)
7 |     return x # new subtree root

```

Remark 4.1. Red-black insertion needs at most 2 rotations; AVL may need $O(\log n)$ up the path. Hence red-black is preferred when insertions dominate, AVL when lookups dominate.

4.5 Chapter Notes

CLRS Ch. 12–13, Weiss Ch. 4.2–4.3. AVL Fibonacci bound proof by induction on height.

4.6 Exercises

- E4.1 **BST ops.** Insert 8, 3, 10, 1, 6, 14, 4, 7 into empty BST in order. Draw the tree after each insertion and list inorder. Delete 3 (two children) and show the successor replacement.
- E4.2 **AVL insertion.** Insert 1, 2, 3, 4, 5, 6 into empty AVL. Show rotations at each step and verify Eq. (4.1) holds.
- E4.3 **Height bound.** Prove $n \geq F_{h+2} - 1$ for AVL of height h (F_k Fibonacci). Deduce $h \leq 1.44 \log_2 n$.
- E4.4 **Red-black.** Insert 10, 20, 30, 15 into empty red-black tree. Show colourings and rotations. Verify all three invariants after each insert.
- E4.5 **Comparison.** Give a sequence where AVL does asymptotically more rotations than red-black on the same insertions, and explain why.

Chapter 5

Heaps & Priority Queues

“The smallest is always on top.” — A heap trades total order for instant access to the extremum.

Prerequisites: Ch. 3 (trees), Ch. 1 (arrays). Heap is an array-embedded tree giving $O(\log n)$ priority-queue ops and $O(n \log n)$ sorting.

Learning Outcomes

After this chapter you will be able to:

- (L1) state the heap invariant and map heap indices to tree positions;
- (L2) implement `push`, `pop`, `heapify` with sift-up/down;
- (L3) build a heap in $O(n)$ via bottom-up heapify and prove it;
- (L4) use priority queues for scheduling and preview heap sort / Dijkstra.

5.1 The Heap Invariant

A (min-)heap is a *complete* binary tree where every node $\leq$ its children. The minimum is at the root. Completeness means all levels are full except possibly the last, filled left-to-right — so the tree maps perfectly to an array.

$$\text{For } i \text{ (1-indexed): } \text{parent}(i) = \lfloor i/2 \rfloor, \text{ left}(i) = 2i, \text{ right}(i) = 2i + 1. \quad (5.1)$$

With 0-indexed arrays: $\text{parent } (i - 1)/2$, children $2i + 1, 2i + 2$.

Not a BST: heap only orders parent vs. child, not left vs. right. Inorder is not sorted.

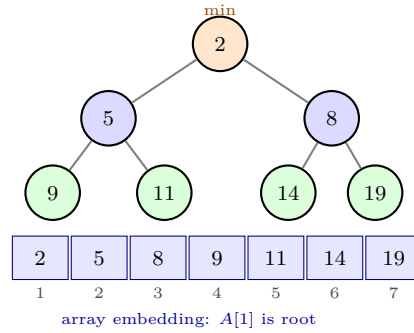


Figure 5.1: Min-heap as tree (top) and array (bottom). Every parent $\leq$ children; completeness gives $O(1)$ index arithmetic.

5.2 Priority-Queue Operations

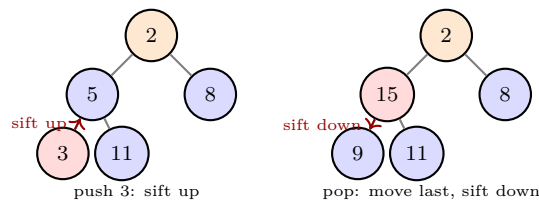


Figure 5.2: Sift-up after insertion (left) and sift-down after extraction (right). Each follows one root-to-leaf path: $O(\log n)$.

```

1  import heapq # Python min-heap (0-indexed)
2
3  def push(heap, x):
4      heap.append(x)
5      i = len(heap) - 1
6      while i > 0:
7          p = (i - 1) // 2
8          if heap[p] <= heap[i]: break
9          heap[p], heap[i] = heap[i], heap[p]
10         i = p # sift up -- O(log n)
11
12 def pop(heap):
13     if len(heap) == 1: return heap.pop()
14     top = heap[0]
15     heap[0] = heap.pop() # move last to root
16     i = 0
17     while True: # sift down -- O(log n)
18         l, r = 2*i+1, 2*i+2
19         smallest = i
20         if l < len(heap) and heap[l] < heap[smallest]: smallest = l
21         if r < len(heap) and heap[r] < heap[smallest]: smallest = r
22         if smallest == i: break
23         heap[i], heap[smallest] = heap[smallest], heap[i]
24         i = smallest
25     return top

```

`peek = heap[0]` in $O(1)$. `decreaseKey` at index i sifts up in $O(\log n)$.

5.3 Building a Heap in $O(n)$

Naively pushing n items costs $O(n \log n)$. Bottom-up **heapify** does it in $O(n)$: process nodes from $n/2$ down to 1, sifting each down. Shorter subtrees dominate the sum.

$$\sum_{h=0}^{\lfloor \log n \rfloor} \lceil n/2^{h+1} \rceil \cdot O(h) = O\left(n \sum_{h \geq 0} h/2^h\right) = O(n), \quad (5.2)$$

since $\sum h/2^h = 2$.

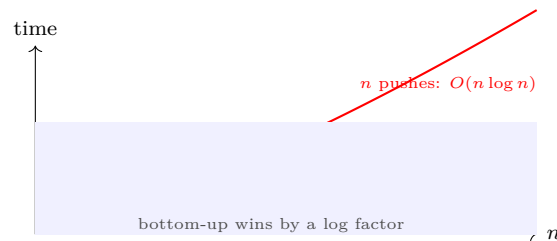


Figure 5.3: Heap construction: repeated push vs. bottom-up heapify.

```

1 // C++ STL priority queue (max-heap by default)
2 #include <queue>
3 std::priority_queue<int> pq;           // max-heap
4 pq.push(5); pq.push(2); pq.push(8);
5 int top = pq.top(); // 8
6 pq.pop();           // removes 8
7
8 // min-heap variant
9 std::priority_queue<int, std::vector<int>, std::greater<int>> minpq;

```

5.4 Applications Preview

Heap sort: build heap $O(n)$, repeatedly pop min $O(n \log n)$ in-place. Dijkstra's algorithm (Ch. 7) uses a min-heap to extract the closest unsettled vertex in $O((n + m) \log n)$.

5.5 Variants & Analysis Details

Max-heap vs. min-heap. Flip comparison. Max-heap gives descending sort by repeated extraction; min-heap is natural for Dijkstra (Ch. 7).

d -ary heap. Each node has d children; height $\log_d n$, sift-down compares d children. Choose d to trade off sift-up vs. sift-down for the workload.

Decrease-key. After lowering $A[i]$, sift up restores heap in $O(\log n)$ — essential for Dijkstra and Prim.

```

1 # heapq is min-heap; max-heap via negation
2 import heapq

```

```

3 max_heap = []
4 for x in [5,2,8,1]: heapq.heappush(max_heap, -x)
5 top = -heapq.heappop(max_heap) # 8
6
7 # Decrease-key at index i (min-heap)
8 def decrease_key(heap, i, new_val):
9     assert new_val <= heap[i]
10    heap[i] = new_val
11    while i > 0:
12        p = (i-1)//2
13        if heap[p] <= heap[i]: break
14        heap[p], heap[i] = heap[i], heap[p]
15        i = p

```

5.6 Chapter Notes

CLRS Ch. 6, Weiss Ch. 6. Heap as priority queue vs. sorted array vs. balanced BST trade-offs.

5.7 Exercises

- E5.1 **Heap ops.** Insert 4, 8, 2, 9, 5, 11, 14, 3 into empty min-heap (array order). Show array after each insert and after two pops.
- E5.2 **Build-heap proof.** Prove Eq. (5.2) sums to $O(n)$ by bounding $\sum h/2^h$. Why does sifting from $n/2$ down to 1 suffice (leaves need no work)?
- E5.3 **Heap vs. BST.** Give an array that is a valid min-heap but whose inorder is not sorted. Explain why heap search for arbitrary key is $\Theta(n)$.
- E5.4 **k -way merge.** Merge k sorted arrays of total size n using a heap in $O(n \log k)$. Prove your bound and show it beats pairwise merging.
- E5.5 **Heap sort in-place.** Describe in-place heap sort on array $A[1..n]$ (build max-heap, swap max to end, shrink). Analyse space $O(1)$ auxiliary and stability.

Chapter 6

Hashing

“Hashing is the closest we get to $O(1)$ search.” — If the hash behaves, the dictionary flies.

Prerequisites: Ch. 1 (arrays), Ch. 5 (analysis). Hashing gives expected $O(1)$ insert/search/delete with high probability.

Learning Outcomes

After this chapter you will be able to:

- (L1) design hash functions and explain universal hashing;
- (L2) implement chaining and analyse $O(1 + \alpha)$ expected cost;
- (L3) implement open addressing (linear/quadratic/double hashing) and analyse clustering;
- (L4) handle load factor, rehashing, and deletion correctly.

6.1 Hash Functions

A hash function $h : U \rightarrow \{0, \dots, m-1\}$ maps universe U into m slots. Ideal: uniform, deterministic, fast. Two classic constructions for integer keys:

$$h(k) = k \bmod m \quad (\text{division, } m \text{ prime away from powers of } 2),$$

$$h(k) = \lfloor m \cdot (kA \bmod 1) \rfloor \quad (\text{multiplication, } 0 < A < 1),$$

$$h(k) = ((ak + b) \bmod p) \bmod m \quad (\text{universal family, } p \text{ prime } > |U|).$$

Universal hashing picks a, b randomly at runtime, guaranteeing low collision probability for any adversarial input set.

Strings: polynomial rolling hash $h(s) = \sum_i s_i \cdot B^i \bmod m$ or Python's salted hash.

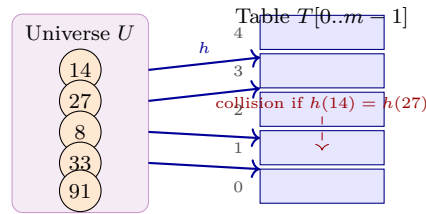


Figure 6.1: Hash function maps many keys into m slots. Collisions are inevitable when $|U| > m$ (pigeonhole) and must be resolved.

6.2 Chaining

Each slot $T[j]$ holds a linked list (chain) of keys hashing to j . Load factor $\alpha = n/m$.

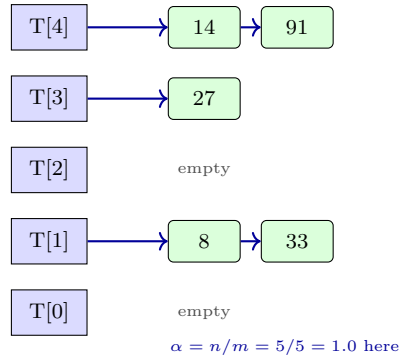


Figure 6.2: Chaining: each slot heads a linked list. Expected chain length = α ; search scans the chain.

Under *simple uniform hashing*, expected search cost is $\Theta(1 + \alpha)$ (hash + chain scan). If $m = \Theta(n)$ then $\alpha = O(1)$ and operations are expected $O(1)$. Worst case $\Theta(n)$ if all keys collide, but universal hashing makes this vanishingly unlikely.

```

1  # Chaining -- Python sketch
2  class ChainedHash:
3      def __init__(self, m=8):
4          self.m = m
5          self.T = [[] for _ in range(m)]
6      def _h(self, k): return hash(k) % self.m
7      def insert(self, k, v):
8          b = self._h(k)
9          for i, (kk, _) in enumerate(self.T[b]):
10             if kk == k: self.T[b][i] = (k, v); return
11             self.T[b].append((k, v))
12      def search(self, k):
13          for kk, v in self.T[self._h(k)]:
14             if kk == k: return v
15          return None

```

Rehashing: when α exceeds threshold (e.g. 0.75), allocate $2m$ slots and reinsert all keys: $O(n)$ but amortised $O(1)$.

6.3 Open Addressing

All keys live in T itself. Probe sequence $h(k, 0), h(k, 1), \dots, h(k, m - 1)$ is a permutation of slots.

- **Linear probing:** $h(k, i) = (h(k) + i) \bmod m$. Suffers *primary clustering* (long runs).
- **Quadratic probing:** $h(k, i) = (h(k) + c_1 i + c_2 i^2) \bmod m$. Reduces primary clustering.
- **Double hashing:** $h(k, i) = (h_1(k) + i \cdot h_2(k)) \bmod m$. Best distribution; $h_2(k) \neq 0$.

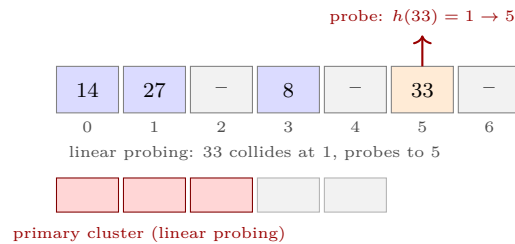


Figure 6.3: Open addressing with linear probing. Collisions probe forward; clusters grow and degrade performance.

Load factor $\alpha = n/m < 1$ (must keep free slots). Expected probes $\approx 1/(1 - \alpha)$ for unsuccessful search with uniform hashing. Deletion cannot just clear a slot (breaks probe chains) — mark as DELETED/tombstone or rehash.

```

1 # Open addressing -- linear probing
2 class OpenHash:
3     EMPTY, DELETED = object(), object()
4     def __init__(self, m=8):
5         self.m, self.T = m, [self.EMPTY]*m
6     def _h(self, k, i): return (hash(k) + i) % self.m
7     def insert(self, k):
8         for i in range(self.m):
9             j = self._h(k, i)
10            if self.T[j] in (self.EMPTY, self.DELETED):
11                self.T[j] = k; return j
12            raise RuntimeError("table full")
13    def search(self, k):
14        for i in range(self.m):
15            j = self._h(k, i)
16            if self.T[j] is self.EMPTY: return None
17            if self.T[j] == k: return j
18        return None

```

6.4 Comparison

6.5 Chapter Notes

CLRS Ch. 11, Weiss Ch. 5. Python dict/set use open addressing; Java HashMap uses chaining with treeification.

Aspect	Chaining	Open addressing
α can exceed 1	yes	no ($\alpha < 1$)
Deletion	easy	tombstones
Cache locality	poor (lists)	excellent
Worst-case search	$\Theta(n)$	$\Theta(m)$

Table 6.1: Chaining vs. open addressing trade-offs.

6.6 Exercises

- E6.1 **Chaining.** $m = 5$, $h(k) = k \bmod 5$, insert 14, 27, 8, 33, 91 with chaining. Draw table and compute expected search cost for present vs. absent keys.
- E6.2 **Linear probing.** Same keys/ m with linear probing. Show probe sequences and final table. Count total probes for all insertions.
- E6.3 **Universal bound.** For universal family $h_{a,b}(k) = ((ak + b) \bmod p) \bmod m$, prove $\Pr[h(k) = h(\ell)] \leq 1/m$ for $k \neq \ell$. Deduce expected chain length $\leq 1 + \alpha$.
- E6.4 **Tombstones.** Show that naively clearing a slot on deletion in open addressing breaks search. Give an example and explain why tombstones fix it but degrade performance over many deletions.
- E6.5 **Load factor.** With $\alpha = 0.9$ and uniform hashing, estimate expected probes for unsuccessful search in open addressing via $1/(1 - \alpha)$. At what α does rehashing become worthwhile?

Chapter 7

Graphs: Representations, Traversals & Preview

“A graph is just vertices and edges.” — Yet it models networks, dependencies, maps, and the web itself.

Prerequisites: Ch. 2 (queues/stacks), Ch. 3 (BFS/DFS intuition), Ch. 5 (priority queue for Dijkstra/MST preview).

Learning Outcomes

After this chapter you will be able to:

- (L1) represent graphs via adjacency matrix vs. adjacency list and compare costs;
- (L2) execute BFS and DFS, classify edges, and analyse $O(n + m)$ time;
- (L3) preview MST (Kruskal/Prim) and Dijkstra’s shortest paths at a high level;
- (L4) model a problem as a graph and choose the right representation.

7.1 Graph Basics & Representations

A graph $G = (V, E)$ has $|V| = n$ vertices and $|E| = m$ edges. Directed vs. undirected; weighted vs. unweighted. Degree $\deg(v)$ counts incident edges.

Two canonical representations:

Implementation: matrix as `vector<vector<int>>` or bitset; list as `vector<vector<int>>` (weighted variant uses `pair<int,int>` for (v, w)).

```
1 # Adjacency list -- Python
2 n = 4
```

Operation	Adj. matrix $n \times n$	Adj. list
Space	$\Theta(n^2)$	$\Theta(n + m)$
Edge (u, v) exists?	$O(1)$	$O(\deg(u))$
List neighbours of u	$\Theta(n)$ scan	$\Theta(\deg(u))$
Sparse $m \ll n^2$	wasteful	efficient
Dense $m \approx n^2$	fine	overhead

Table 7.1: Adjacency matrix vs. list. Choose list for sparse graphs (most real networks).

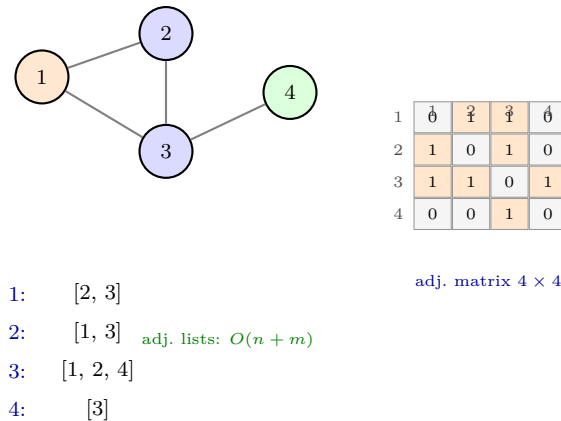


Figure 7.1: Undirected graph with 4 vertices, 4 edges: adjacency matrix (symmetric, orange = 1) vs. adjacency lists.

```

3 adj = [[] for _ in range(n)]
4 def add_edge(u, v, w=1): # 0-indexed, undirected
5     adj[u].append((v, w))
6     adj[v].append((u, w))
7 add_edge(0,1); add_edge(0,2); add_edge(1,2); add_edge(2,3)
8 # adj[2] == [(0,1),(1,1),(3,1)]

```

7.2 BFS & DFS

Both traverse in $O(n + m)$ with adjacency lists ($O(n^2)$ with matrix).

BFS uses a queue: explores in increasing distance from source. Yields shortest paths (fewest edges) in unweighted graphs. Records **dist** and **parent**.

DFS uses a stack (recursion): goes deep before backtracking. Classifies edges as tree/back/forward/cross (directed case). Useful for topological sort and cycle detection.

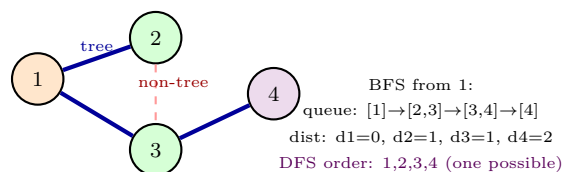


Figure 7.2: BFS tree from vertex 1 (blue thick = tree edges). Dashed edge (2,3) is non-tree; BFS finds shortest-path distances.

```

1 from collections import deque

```

```

2  def bfs(adj, s):
3      n = len(adj)
4      dist = [-1]*n; parent = [-1]*n
5      dist[s]=0; q=deque([s])
6      order=[]
7      while q:
8          u=q.popleft(); order.append(u)
9          for v,_ in adj[u]:
10             if dist[v]==-1:
11                 dist[v]=dist[u]+1; parent[v]=u; q.append(v)
12     return order, dist, parent
13
14 def dfs(adj, s, visited=None, order=None):
15     if visited is None: visited=set(); order=[]
16     visited.add(s); order.append(s)
17     for v,_ in adj[s]:
18         if v not in visited: dfs(adj, v, visited, order)
19     return order

```

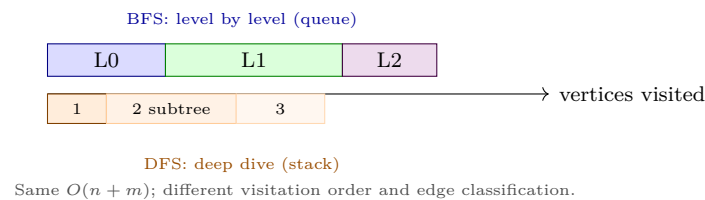


Figure 7.3: BFS vs. DFS visitation pattern on the same graph. BFS is level-order; DFS plunges deep.

7.3 Preview: MST & Shortest Paths

Minimum Spanning Tree (MST): cheapest set of $n - 1$ edges connecting all vertices. Kruskal sorts edges and union-finds; Prim grows via min-heap (Ch. 5): both $O(m \log n)$.

Dijkstra: single-source shortest paths for non-negative weights. Repeatedly extracts min-distance vertex from a heap and relaxes edges: $O((n + m) \log n)$ with binary heap. BFS is the special case $w = 1$.

These are developed fully in SC2001; here we note that the right heap + graph representation gives the right bound.

7.4 Chapter Notes

CLRS Ch. 20–24, Weiss Ch. 9. Adj. list is default for sparse graphs; matrix for dense or fast edge queries. BFS/DFS are the workhorses for connectivity, bipartiteness, and topological sort.

7.5 Exercises

- E7.1 **Representations.** For $n = 1000$, $m = 3000$, compute space for matrix (1 byte/entry) vs. list (8 bytes/edge endpoint). At what m does matrix win?
- E7.2 **BFS distances.** Run BFS from vertex 1 on Fig. 7.1. Give `dist`, `parent`, and BFS tree. Prove BFS indeed gives shortest-path distances in unweighted graphs.
- E7.3 **DFS classification.** Run DFS from 1 on the same graph (neighbour order increasing). Classify each edge as tree/back/forward/cross (treat undirected edges as two directed arcs).
- E7.4 **Dijkstra preview.** Add weights $w(1, 2) = 4, w(1, 3) = 2, w(2, 3) = 1, w(3, 4) = 5$. Run Dijkstra from 1 by hand with a heap; show extract order and final distances. Why does negative weight break it?
- E7.5 **MST preview.** Give MST of the weighted graph above (list edges and total weight). Run Kruskal and Prim and compare steps.

Chapter 8

Sorting & Asymptotic Analysis

“Sorting is the most studied problem in computing.” — And the sharpest lens for $O(\cdot)$ reasoning.

Prerequisites: Ch. 1–7 (arrays, heaps, recurrences, $O(\cdot)$). This chapter ties SC1007 together: classic sorts, linear-time sorts, lower bounds, and recurrence preview.

Learning Outcomes

After this chapter you will be able to:

- (L1) implement and analyse quicksort, merge sort, and heap sort;
- (L2) apply counting sort and radix sort for $O(n)$ sorting under assumptions;
- (L3) prove the $\Omega(n \log n)$ comparison lower bound via decision trees;
- (L4) solve recurrences and compare $O(n \log n)$ vs. $O(n^2)$ vs. $O(n)$.

8.1 Comparison Sorts: Quick, Merge, Heap

All comparison sorts must be $\Omega(n \log n)$ worst-case (decision-tree bound, Sec. 8.3). Three $O(n \log n)$ champions:

Merge sort: divide at mid, recurse, merge $\Theta(n)$. Recurrence $T(n) = 2T(n/2) + \Theta(n) \Rightarrow T(n) = \Theta(n \log n)$ worst-case. Stable, not in-place (needs $\Theta(n)$ auxiliary).

Heap sort: build max-heap $O(n)$, repeatedly extract max $O(n \log n)$ in-place, not stable.

Quicksort: partition around pivot $\Theta(n)$, recurse on sides. Average $\Theta(n \log n)$; worst $\Theta(n^2)$ (bad pivots) but randomised pivot gives expected $\Theta(n \log n)$. In-place, fast in practice.

```
1 # Quicksort -- Lomuto partition, randomised pivot
2 import random
3 def partition(a, lo, hi):
4     pivot = a[hi]
```

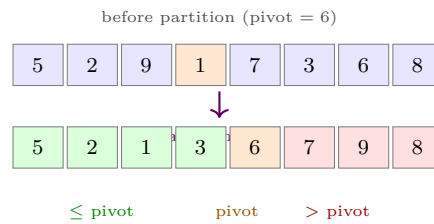


Figure 8.1: Quicksort partition: pivot 6 moves to final position; left $\leq$ pivot, right $>$ pivot. Recurse on both sides.

```

5 |     i = lo
6 |     for j in range(lo, hi):
7 |         if a[j] <= pivot:
8 |             a[i], a[j] = a[j], a[i]; i += 1
9 |     a[i], a[hi] = a[hi], a[i]
10 |    return i
11 |
12 | def quicksort(a, lo=0, hi=None):
13 |     if hi is None: hi = len(a)-1
14 |     if lo < hi:
15 |         # random pivot for expected O(n log n)
16 |         r = random.randint(lo, hi); a[r], a[hi] = a[hi], a[r]
17 |         p = partition(a, lo, hi)
18 |         quicksort(a, lo, p-1); quicksort(a, p+1, hi)

```

```

1 | # Merge sort -- stable, O(n log n) worst-case
2 | def merge(a, lo, mid, hi, tmp):
3 |     i, j, k = lo, mid+1, lo
4 |     while i <= mid and j <= hi:
5 |         if a[i] <= a[j]: tmp[k]=a[i]; i+=1
6 |         else: tmp[k]=a[j]; j+=1
7 |         k+=1
8 |     while i <= mid: tmp[k]=a[i]; i+=1; k+=1
9 |     while j <= hi: tmp[k]=a[j]; j+=1; k+=1
10 |    a[lo:hi+1]=tmp[lo:hi+1]
11 |
12 | def mergesort(a, lo=0, hi=None, tmp=None):
13 |     if hi is None: hi=len(a)-1; tmp=[0]*len(a)
14 |     if lo < hi:
15 |         mid=(lo+hi)//2
16 |         mergesort(a, lo, mid, tmp); mergesort(a, mid+1, hi, tmp)
17 |         merge(a, lo, mid, hi, tmp)

```

8.2 Linear-Time Sorts

When keys have structure, we beat $n \log n$.

Counting sort: for integers in $[0, k]$, count frequencies, prefix sums give positions. Time $\Theta(n + k)$, space $\Theta(k)$. Stable.

Radix sort: sort by digit from LSD to MSD using stable counting sort per digit. For d digits base b :

$O(d(n+b))$.

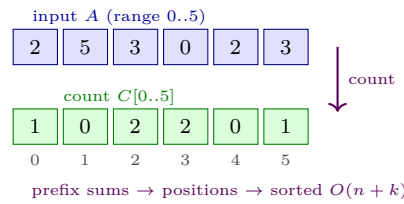


Figure 8.2: Counting sort: histogram C then prefix sums place each element directly. Radix sort repeats this per digit.

```

1  # Counting sort --  $O(n+k)$ , stable
2  def counting_sort(a, k):
3      c = [0]*(k+1)
4      for x in a: c[x] += 1
5      for i in range(1, k+1): c[i] += c[i-1]
6      out = [0]*len(a)
7      for x in reversed(a): # reverse for stability
8          c[x]-=1; out[c[x]]=x
9      return out
10
11 # Radix sort -- LSD first, stable per digit
12 def radix_sort(a, base=10):
13     maxv = max(a) if a else 0
14     exp = 1
15     while maxv // exp > 0:
16         # counting sort by digit (exp place)
17         buckets = [[] for _ in range(base)]
18         for x in a: buckets[(x//exp) % base].append(x)
19         a = [x for b in buckets for x in b]
20         exp *= base
21     return a

```

8.3 Lower Bound & Analysis

Theorem 8.1 (Comparison lower bound). Any comparison sort needs $\Omega(n \log n)$ comparisons worst-case.

Sketch. Decision tree has $n!$ leaves (one per permutation). Binary tree with L leaves has height $\geq \lceil \log_2 L \rceil$. So height $\geq \log_2(n!) = \Theta(n \log n)$ by Stirling. $\square$

Recurrence preview: merge sort $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \log n)$; quicksort average $T(n) = T(k) + T(n-k-1) + \Theta(n)$ with $k \sim \text{Uniform}$ gives $\Theta(n \log n)$ expected.

8.4 Stability, Hybrids & Recurrence Details

Stability. A sort is *stable* if equal keys retain input order. Merge and counting/radix are stable; heap and quicksort (as usually implemented) are not. Stability matters when sorting by multiple keys.

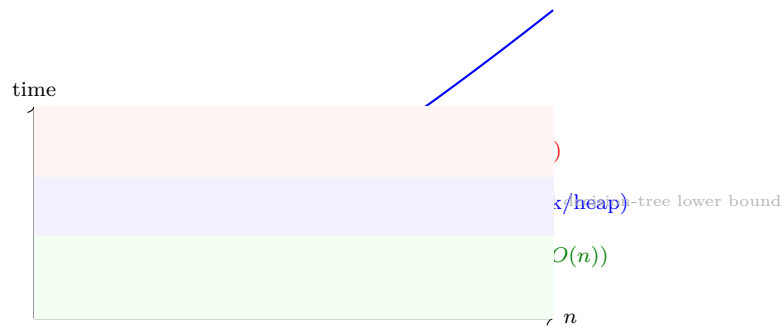


Figure 8.3: Sorting landscape: comparison sorts cannot beat $n \log n$; counting/radix break the bound by using key structure.

Introsort (STL `sort`) runs quicksort but switches to heapsort when recursion depth exceeds $2 \log n$, guaranteeing $O(n \log n)$ worst-case while keeping quicksort’s speed. Small subarrays (< 16) use insertion sort.

Recurrence bridge to SC2001. Merge sort gives $T(n) = 2T(n/2) + \Theta(n)$. Master theorem (SC2001 Ch. 3) yields $\Theta(n \log n)$. Quicksort average case uses indicator variables: expected comparisons $= 2 \sum_{i < j} 1/(j-i+1) = \Theta(n \log n)$.

```

1  # Introsort sketch -- depth-limited quicksort
2  import heapq
3  def introsort(a, lo, hi, depth_limit):
4      if hi - lo < 16:
5          insertion_sort(a, lo, hi); return
6      if depth_limit == 0:
7          heapsort_range(a, lo, hi); return
8      p = partition(a, lo, hi)
9      introsort(a, lo, p-1, depth_limit-1)
10     introsort(a, p+1, hi, depth_limit-1)

```

8.5 Chapter Notes

CLRS Ch. 2, 6–8, Weiss Ch. 7. Quicksort is usually fastest in practice despite worst case; introsort (STL `sort`) switches to heapsort when recursion deepens.

8.6 Exercises

E8.1 **Quicksort trace.** Sort $[5, 2, 9, 1, 7, 3, 6, 8]$ with quicksort, pivot = last element. Show partitions and recursion tree. Count comparisons.

E8.2 **Merge analysis.** Prove $T(n) = 2T(n/2) + n = \Theta(n \log n)$ by recursion tree. Is merge sort stable? Is it in-place? Justify.

E8.3 **Counting sort.** Sort $[2, 5, 3, 0, 2, 3]$ with counting sort ($k = 5$). Show C , prefix sums, and output construction. Why does iterating reversed preserve stability?

E8.4 **Lower bound.** Prove $\log_2(n!) = \Theta(n \log n)$ using Stirling or integral bounds. Explain why counting sort

does not violate the comparison lower bound.

E8.5 Hybrids. Describe introsort (quicksort $\rightarrow$ heapsort fallback + insertion sort for small n). Why does STL use it? Analyse its $O(n \log n)$ worst-case guarantee.